

LI Linghao

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EDUCATION

Zhejiang University of Technology (ZUT)

Bachelor of Management in Information Management and Information Systems

GPA: 3.78 / 4.00

Coursework: Data Structures; Database Principles A; Applied Statistics A; Large Data Mining and Machine Learning; Business Intelligence

Hangzhou, China

Sep 2021 – Jun 2025

PUBLICATIONS

First Author, “Fault Early Warning Model for Wind Turbines Based on Imbalanced Positive and Negative Sample Contrastive Self-Supervised Learning”, Computer Integrated Manufacturing Systems, online first, Sep 2025, pp. 1 – 18, DOI: 10.13196/j.cims.2025.0055.

- Developed a four-module pipeline for highly imbalanced SCADA data (boxplot-based downsampling, kernel Bayesian VAR for small-sample faults, attention VAE for normal-state modelling, adaptive threshold tree for decision boundaries), validated on 24 months of data from 6 turbines (34 features, 367 fault types), outperforming SVM, LightGBM, LSTM and vanilla autoencoders in fault early warning and supporting cost reduction and downtime mitigation in wind farm operations.

INTERNSHIP EXPERIENCE

Shanghai Boyin Leading Education Technology Co., Ltd.

AI Product Manager Intern

Shanghai, China

Jun 2025 – Nov 2025

- Participated in end-to-end product planning for the smart campus portal and campus information system, decomposing course selection, academic notifications and approval flows into AI use cases and drafting PRDs with user flows and data models to drive iterative delivery.
- Designed conversation flows for “Campus Intelligent Assistant” and “Smart Service Hall” using LLM and RAG, defining knowledge base schemas, vector search fields and data sync rules to reduce manual handling of high-frequency academic and enrolment queries.
- Built the key metrics framework for smart campus modules, contributing to event tracking design and log specifications, and used Python and SQL to analyse traffic, conversion funnels, retention and response time to generate data-driven UI and journey optimisations.
- Conducted in-depth interviews and usability tests with students and staff, consolidating pain points and behavioural data into a prioritised product backlog, and worked with the IT centre to implement SSO, role-based access control and API standards for stable AI integration.

Zhejiang Torch Productivity Promotion Center Co., Ltd.

Data Analyst, R&D Department

Hangzhou, China

Feb 2025 – May 2025

- Designed and implemented a dynamic programming – based large scale order consolidation model, explicitly modeling time windows, destinations and weight constraints to improve shipment batching and vehicle load factor on historical data.
- Jointly optimised consolidation strategy and tiered freight pricing rules, tuning group weights to push more batches into lower price tiers and reducing simulated total transport cost by 5.84% (about 3.1 billion pricing units).
- Built an evaluation framework covering unit price, cost per ton-kilometre, average load factor and consolidation rate, and ran scenario analyses across time windows and rules to select parameters for production deployment.
- Designed a Gantt chart – style visualisation of consolidated orders, showing dispatch windows and relative weights per batch to help business and management teams inspect strategy effects and identify further optimisation opportunities.

Guoxin Big Data Application Research Institute

Data Analyst, Industrial Applications Department

Hangzhou, China

Jun 2023 – Sep 2023

- Led development of core prediction algorithms for handheld wind turbine inspection devices, proposing a BiScaleWaveNet dual stream heterogeneous neural network in Python and PyTorch for end to end modelling of gearbox oil hyperspectral sequences.
- Introduced a learnable wavelet transform module to decompose raw spectra into high frequency transient and low frequency trend features, mitigating band mixing and significantly improving feature representation and prediction stability over multiple CNN and LSTM baselines.
- Built a multi element quantitative prediction framework, using cross validation and metrics such as MAE and R^2 to evaluate architectures and hyperparameters, and delivered an engineering ready solution to the multi element prediction problem for gearbox oil.
- Collaborated with hardware and field engineering teams to co tune the system, constraining inference latency and memory footprint within handheld limits and mapping continuous predictions into alert levels via threshold rules to support fast on site diagnostics.

RESEARCH & PROJECT EXPERIENCE

Yidingtang Software Technology Co., Ltd. Hangzhou Branch

Sole Developer, Virtual Simulation Intelligent Procurement Data Analysis Educational Software

Mar 2025 – Jun 2025

- Designed and implemented a machine learning analytics module for virtual procurement teaching scenarios, building a pipeline from data preprocessing and feature engineering to model training and evaluation to enable configurable intelligent decision making.
- Built a feature engineering framework using filter methods with chi square tests and correlation analysis, embedded methods using tree based feature importance, and wrapper methods with recursive feature elimination to automate feature selection and reduce overfitting.
- Developed visual training components for decision tree and feedforward neural network models, dynamically displaying tree growth, node split rules, and loss and accuracy curves across iterations to make algorithms and training behaviour transparent to instructors and students.
- Conducted model comparison experiments using k fold cross validation and grid search, evaluating different feature sets and parameter configurations by explainability, training time and accuracy to derive recommended configurations for classroom demonstrations.
- Implemented modular machine learning scripts in Python with pandas, NumPy and scikit learn, encapsulating data processing, feature selection, model training and visualisation into reusable function interfaces for rapid reuse in procurement cases and teaching projects.

Wenzhou Weijiguang Intelligent Technology Co., Ltd.

Technical Consultant, AI-Assisted Tender Document Writing LLM Program

Jan 2025 – Feb 2025

- Architected the AI tender-assistance system with a RAG-based design, building vectorized retrieval and indexing over historical tenders, case libraries and qualification files to unify access to structured and unstructured data.
- Built an LLM-driven retrieve–recompose–generate pipeline, using text embeddings and similarity search to select high-relevance content and injecting tender clauses and scoring criteria into prompt templates to control coverage of key indicators and compliance items.
- Led LLM finetuning and prompt-engineering iterations, curating typical tender paragraphs and FAQ samples for training and evaluation and defining quality rules on logical coherence, content completeness and compliance risk.
- Developed core application flows in Python using LangChain, encapsulating retrieval, question answering and template-filling chains as reusable modules that call OpenAI APIs for generation, rewriting, refinement and consistency checking.
- Defined deployment guidelines and usage policies, adding sensitive-field masking, source citation and mandatory human review so that the LLM component could be embedded into existing bidding workflows and scaled to additional business lines.

Zhejiang Wenzhou Southern Suburb Yundian Garment Factory

Technical Consultant, AIGC-Powered Fashion Design Sketch Generation Program

Jan 2025 – Feb 2025

- Led the AIGC technical roadmap for women's sportswear sketch generation, decomposing business needs into data and model pipelines and defining an end-to-end architecture from design feature encoding and conditional control to sketch generation and feedback.
- Built a proprietary sketch and style knowledge base by tagging historical designs, pattern structures, stitching and panel details and color schemes, and constructing label schemas and feature vectors to support high quality retrieval and generation.
- Encoded style, product category and sports scenario as conditional variables based on metadata and style tags, injecting them into generation constraints so that the model could produce multiple candidate sketch options while preserving brand tone and improving exploration efficiency.
- Used Python and LangChain to integrate OpenAI APIs and designed a GraphRAG-inspired pipeline that combines graph based structure and vector retrieval, explicitly modeling relationships between design elements to drive prompt construction and result reordering.
- Modularized requirement parsing, reference image retrieval, style element extraction and sketch description generation as configurable components with parameter settings and logging to enable rapid reuse and systematic evaluation in new collections and later iterations.

Library of Zhejiang University of Technology

Lead Developer, LLM-Based Automated Academic Paper Knowledge Network Construction Platform

Mar 2025 – Jun 2025

- Designed and implemented the overall architecture of an LLM-based literature analysis system, linking retrieval, text parsing, topic extraction and LLM reasoning into a single pipeline to process batches of papers and generate domain research maps.
- Built a multi-query knowledge network construction mechanism that combines keyword matching, semantic similarity and co-occurrence to form a two-layer “paper–concept” graph, revealing research directions and cross-cutting themes beyond traditional single-point search.
- Developed a review generation strategy driven by the knowledge graph, encoding key nodes and their neighbourhoods into LLM prompts so that the model produces multi-perspective, structured literature reviews and significantly cuts the time to grasp a new field.
- Implemented a full-stack application using Vue3 for the interactive front end and Python FastAPI for backend services and async tasks, integrating Alibaba Qwen LLM for semantic analysis and information extraction to support end-to-end retrieval, graph exploration and review generation in one interface.

Knowledge Management Interconnected Platform Full-Stack Program

Sole Developer

Jul 2024 – Sep 2024

- Designed and implemented backend data models and API layer for heterogeneous PDF, Markdown and knowledge graph content, unifying document metadata, sections and graph relations to enable fine grained cross format linking, full traceability and consistent access control.
- Built an interactive knowledge graph editor in Vue 3 with a graph visualization library, supporting node and edge creation, deletion, property editing and drag and drop layout, and optimized rendering with virtual lists for smooth interaction over thousands of nodes.
- Integrated file management, note taking and graph visualization into a single workspace, creating a document–note–graph knowledge loop that converts scattered files into searchable, recomposable knowledge assets and reduces team rediscovery cost.
- Used Neo4j as the graph store and designed advanced Cypher query templates for path finding, pattern matching and statistical analysis, while building a FastAPI and Vue 3 full stack framework to orchestrate backend services and front end visualization.

Deep Learning Model End-to-End Research-Oriented Experimental Platform Full-Stack Program

Sole Developer

Jul 2024 – Sep 2024

- Led the design of a full stack experiment platform for modular deep learning models, abstracting dataset definitions, model architecture descriptions, training workflows and result dashboards into a unified metadata driven pipeline with end to end traceability.
- Developed a visual model composer that wraps convolution layers, fully connected layers, regularization blocks and loss functions as drag and drop components so that researchers can quickly assemble, save and reuse alternative network designs.
- Standardized the training lifecycle by unifying logging, key metrics and model version records, providing a common foundation for large scale hyperparameter search and architecture tuning and improving the organization of comparative experiments.
- Implemented custom message queue, cache and thread pool modules in Python to schedule training jobs asynchronously and manage resources, and built a FastAPI plus Vue 3 stack to keep front end configuration, backend training and result feedback in sync.

Technical Research on the Intelligentization of Power Material Management Supported by LLM Two-Layer Nesting

Lead Architect, State Grid Zhejiang Electric Power Co., Ltd.

May 2024 – Jun 2024

- Planned the technical roadmap for power material management, proposing a 2 layer nesting method for secondary learning on large language models that aligns general capabilities with domain specific corpora and strengthens understanding and reasoning on material terminology, regulations and workflows.
- Designed four key methods around material codes, supplier information and equipment lineage data, covering material genealogy generation, demand knowledge network construction, scenario specific application path generation and secondary learning to connect data governance, knowledge modeling and intelligent applications.

- Produced the end to end technical blueprint and module breakdown, defining responsibilities and interfaces for data, knowledge and model layers and providing an executable plan for subsequent algorithm development, system integration and deployment.
- State Grid Project Planning Intelligent Measurement Backend**
- Phase Two Lead Backend Developer, State Grid Zhejiang Electric Power Co., Ltd. Jul 2023 – Aug 2024
- Led phase two backend architecture refactor and core module rewrite, introducing layered design and asynchronous processing to decouple critical logic and significantly improve stability and horizontal scalability in high volume estimation scenarios.
 - Built an intelligent parser for Excel based project estimations and Word feasibility reports, combining regular expressions, rule engines and natural language processing to key figures, clauses and technical parameters and load them into structured storage with high accuracy.
 - Constructed a project corpus aligned with business rules and implemented graphical configuration and dynamic parsing of Excel factor formulas, encapsulating complex estimation logic as visual factor models that business users can adjust, which increases adaptability across project types.
 - Used Python to identify and fix legacy defects, add exception handling and unit tests and refactor and optimize the performance of key modules, then delivered multiple new business features while preserving compatibility and maintainability.
- State Grid Scientific and Technological Resources Search Engine Backend**
- Lead Backend Developer, State Grid Zhejiang Electric Power Co., Ltd. Jun 2023 – Oct 2024
- Led end to end backend design and development for an enterprise heterogeneous knowledge graph platform, using a graph database as a unified knowledge base to align entities and relationships across patents, papers, projects and personnel.
 - Designed and implemented a dual engine search architecture that combines relational and graph based retrieval, supporting efficient keyword and filter queries as well as graph based association and path reasoning to serve different levels of knowledge discovery.
 - Developed advanced capabilities for graph data reconstruction, key term ranking and technical term semantic summarization, leveraging node importance, text statistics and semantic similarity to monitor structural evolution and identify emerging themes.
 - Implemented a Neo4j based graph retrieval engine and a MySQL based structured retrieval engine in Python and integrated them through an intermediate routing and result fusion layer that handles ranking and deduplication and exposes a unified, extensible search service for multiple business systems.

VOLUNTEERING EXPERIENCE

- Opening Ceremony, “Youth Cup” May Fourth Torch Relay, Youth Volunteers Association (YVA), ZUT** 04 May 2023
- Coordinated route planning, materials and venue setup, then served as front line marshal guiding 500+ attendees and supporting athlete check in, torch handovers, rehearsals and contingency drills, helping deliver a zero error opening ceremony.
- Volunteer Base at Yue Temple (Lian Heng Memorial Hall), Xihu Admin., Information Chapter, YVA, ZUT** 01 Oct 2022
- Provided guided tours for 200+ visitors, combining historical narrative and exhibit interpretation to communicate cross strait cultural exchange, while maintaining order, protecting artefacts and handling around 50 daily information requests.
- Orientation Volunteer Service, School of Management, YVA, ZUT** 27 Aug 2022 – 28 Aug 2022
- Assisted over 300 freshmen and parents through full registration, logistics and campus navigation, distributing 150+ welcome packs with zero errors and cutting average check in time by an estimated 20%.
- Environmental Protection Campaign at Jingshan Cuigu, Chemical Engineering Chapter, YVA, ZUT** 28 May 2022
- Led a 5 member team in a 3 km eco clean up and public outreach, delivering environmental education to 100+ visitors and collecting more than 35 kg of sorted waste to improve the local ecosystem.
- Rare Disease Awareness and Support Campaign, Environmental Chapter, YVA, ZUT** 25 Feb 2022 – 27 Feb 2022
- Co organized a rare disease advocacy programme by producing educational materials reaching 1,000+ people, supporting 20+ patients and families with one to one activities, and helping run three public talks with 200+ participants, followed by a 5,000 word impact report..

AWARDS

- Awardee, “Artificial Intelligence+” Track**, 19th Zhejiang Province “Challenge Cup” Extracurricular Academic Science and Technology Competition for College Students Jul 2025
- Successful Participant**, Mathematical Contest in Modeling (MCM) May 2024
- Finalist**, 2023 Certificate Authority Cup International Mathematical Contest in Modeling Jan 2024
- Third-Class Academic Scholarship** Jun 2022

LANGUAGE & SKILL & CERTIFICATE

Languages

- English: IELTS 6.5; CET-4 637; CET-6 570
- Chinese: Native

Technical Skills

- Programming: Python (advanced), Java, R, MATLAB
- AI and Data: deep learning model design, large language model application development, machine learning, data analysis
- Web and Systems: Python backend development (for example FastAPI), Vue based frontend development, full stack prototyping
- Databases: relational databases, Neo4j graph database, data modeling and query design

Certificates

- Neo4j Certified Professional, Neo4j GraphAcademy, Aug 2023